Compact *I–V* Model for Double-Gated MoS_2 FETs Including Short-Channel Effects

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***Abstract*— This article presents a physics-based analytical compact model for double-gated molybdenum disulfide (MoS_2) field effect transistors (FETs), incorporating key physical and short-channel effects (SCEs), such as mobility degradation and velocity saturation. The model is developed from a unified charge control model by evaluating the charge density within the 2-D MoS_2 layer, represented using the Lambert W function, which provides an analytical expression valid and continuous from the subthreshold to the above threshold regime. The drain current is then derived from this unified charge control model, and as a function of closed-form equations for the charge densities at the source and drain ends of the channel. Despite its simplicity, the model shows excellent agreement with experimental data for channel lengths down to 60 nm, making it a powerful tool for accurately predicting the performance of downscaled devices. By including SCEs, this work extends previous modeling efforts and provides a more comprehensive framework for the simulation and optimization of 2-D material-based FETs in circuit design.**

***Index Terms*— 2-D field effect transistors (FETs), 2-D materials, dc compact model, Lambert W function, mobility degradation, molybdenum disulfide (MoS_2) FETs, short-channel effects (SCEs), velocity saturation.**

I. Introduction

TWO-DIMENSIONAL materials have recently emerged as highly promising alternatives for conducting channels in field effect transistors (FETs). This transition is driven by the need to overcome the performance bottlenecks associated with traditional silicon-based CMOS devices, especially as these limitations become more pronounced in very large scale integration (VLSI) circuits [1]. The shrinking of silicon technology, governed by Moore's law, has led to increased challenges such as short-channel effects (SCEs), power dissipation, and parasitic capacitances, which degrade device performance. In contrast, 2-D materials, with their atomic-scale thickness, offer a unique advantage: they provide excellent electrostatic gate control over the channel, which is crucial for mitigating SCEs and minimizing parasitic losses. This ultrathin nature enhances the scalability of FETs and enables the fabrication of devices with higher packing densities than conventional 3-D gate structures. Moreover, 2-D materials hold substantial potential for integration into advanced transistor architectures, such as stacked nanosheet transistors, which represent a future direction for post-silicon electronics [2]. These materials provide a versatile platform for designing new device architectures that can further push the boundaries of miniaturization and performance. Among the diverse range of 2-D materials, transition metal dichalcogenides (TMDs) have attracted considerable attention due to their wide range of electronic properties, including insulating, semiconducting, metallic, and even superconducting behaviors [3]. TMDs, such as molybdenum disulfide (MoS_2), have been extensively studied and are among the most promising candidates for next-generation FETs. High-performance MoS_2 FETs have been demonstrated, showcasing reassuring electrical characteristics such as high ON/OFF ratios and high mobility [4], [5] [6]. However, the most challenging aspects of using those devices in mass production are to achieve low contact resistances and low densities of interface traps [7].

For any emerging technology to find application in commercial electronic systems, particularly in integrated circuits, the development of accurate and reliable compact models is essential. Without such models, the integration of new devices, like 2-D FETs, into VLSI circuits would be impractical. To this end, several compact models for 2-D FETs have been developed in recent years [8], [9] [10], [11] [12], [13] [14], [15], [16] [17]. Cao et al. [8] developed a model that uses a numerical approximation to obtain the current equation. In addition, Taur et al. [9] used a numerical procedure to form the all-region current model. The models developed in [10] and [11] lack the dynamic regime operation. Specifically, the model developed by Suryavanshi and Pop [16] provides a calculation for the drain current considering the contact resistance, traps and impurities, quantum capacitance, fringing fields, high-field velocity saturation, and self-heating. However, in the dynamic regime, this model is based on a charge

Received 26 June 2025; revised 5 September 2025 and 6 October 2025; accepted 8 October 2025. Date of publication 14 November 2025; date of current version 10 December 2025. This work was supported in part by European Union Bayesian inference with flexible electronics for biomedical applications (BAYFLEX) under Contract 101099555 and in part by the Ministry of Science of Spain under Contract PID2021-122399OB-I00. The review of this article was arranged by Editor M. A. Pavanello. *(Corresponding author: Benjamin Iñiguez.)*

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Digital Object Identifier 10.1109/TED.2025.3622099

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model for bulk MOSFETs, which assumes that the channel is always in the weak-inversion regime and that the channel charge is linearly distributed along the device. This assumption is not justified for 2-D devices that operate in accumulation mode in the ON-state [18]. In addition, the model suffers from convergence problems when incorporated into SPICE circuit simulators.

In this article, we present an I–V compact model for short-channel 2-D FETs. This model is based on our prior work [14], [15], where we introduced a dc compact model tailored specifically for long-channel 2-D FETs. The preliminary model focused on capturing the core I–V characteristics of long-channel 2-D FETs, providing a foundational framework for circuit simulation. In this extended version, we enhance the model by incorporating critical physical and SCEs, such as mobility degradation, velocity saturation, channel length modulation (CLM), and drain-induced barrier lowering (DIBL), which are essential for accurately predicting the behavior of scaled-down devices. These enhancements are vital for improving the accuracy and applicability of the model in simulating 2-D FETs.

Our model is based on a unified charge control approach, which is formulated [14], [15] by extending the charge control concepts initially outlined in [18]. One of the features in our model is the inclusion of a closed-form approximation of the channel charge sheet density, represented using the Lambert W function—a mathematical function that is well-established and supported in most circuit simulators. The resulting expression is valid and continuous from the subthreshold to the above threshold regime. The current expression in our model is derived from the unified charge control model and written in terms of charge sheet densities at the source and drain ends of the channel, providing a closed-form solution that can be easily implemented into standard circuit simulation tools, unlike many existing models that rely on separate expressions for different operating regimes, making implementation significantly more tedious. This closed-form nature ensures computational efficiency during simulations. Additionally, the simplicity of the model allows circuit designers to perform primary hand calculations, offering a straightforward way to estimate key device parameters and performance metrics without relying solely on complex numerical simulations. This feature is particularly valuable in the early stages of circuit design, where quick, approximate calculations can guide design decisions and facilitate rapid prototyping.

We have successfully validated our model against experimental data for devices fabricated at Intel Foundry Technology Research Labs and measured at TU Vienna, demonstrating very good agreement across various operating conditions for channel lengths down to the shortest available length (60 nm). The ability of our model to match experimental results demonstrates its potential utility in aiding the design and optimization of circuits that use 2-D FET technology.

This article is structured as follows. In Section II, we present a review of the formulation for the model presented in [14] and the development of our short-channel devices compact dc model. In Section III, we discuss the results and the validation of our model against experimental data obtained from MoS_2 transistors with a channel width of 1 μm and channel lengths of 500, 200, and 60 nm. Finally, Section IV summarizes the main contributions of our model.

II. Model Development

A. Core Model Development

In developing our core model, we start by modeling the electrostatics of the device. Our work is based on [14], [15], [18]. We consider a single-gate operation, with the top gate biased at V_g and the back gate grounded ($V_b = 0$), so (1) describes the electrostatics for top gate control with C_t as follows:

$$Q_{\text{net}} + Q_{\text{it}} = -C_t \left(V_g - V_{g0} - V(x) + V_c(x)\right) \tag{1}$$

where Q_{net} is the total mobile charge density, Q_{it} is the interface trap charge density, C_t is the oxide capacitance per unit area $C_t = \epsilon_0\epsilon_t/t_t$, and $(V_g - V_{g0})$ is the overdrive voltage, where V_{g0} takes into account the work function difference and any possible additional charges due to impurities or doping. The interface trap charge density Q_{it} is significant in MoS_2 FET devices. However, we ignore the dependence of Q_{it} on the gate voltage as a first approximation. This does not affect the physical meaning of the expressions as we consider Q_{it} as a constant value that affects the threshold voltage V_{g0} and the subthreshold slope SS. We do not attempt to capture the dynamics of trapping/de-trapping or bias history dependence. As a result, sweep-direction and rate-dependent hysteresis are not explicitly reproduced and are left for future work.

$V(x)$ is the quasi-Fermi level along the channel, where $V(x = 0) = V_s$ (source voltage) and $V(x = L) = V_D$ (drain voltage). $V_c(x)$ is the shift of the quasi-Fermi level with respect to the conduction band.

Another equation for the overall charge density is introduced to express the charge in terms of quantum capacitance, which will be used to construct the unified charge control model

$$Q_{\text{net}} = -qn(x) = -C_{\text{dq}} V_{\text{th}} \ln\left(1 + \mathrm{e}^{\frac{-V_c(x)}{V_{\text{th}}}}\right) \tag{2}$$

where $C_{\text{dq}} = q^2 - D_0$ is defined as the degenerated-quantum capacitance and D_0 is defined as the density of states and is expressed as follows:

$$D_0 = g_K \left(\frac{m^K}{2\pi\hbar^2}\right) + g_Q \left(\frac{m^Q}{2\pi\hbar^2}\right) \exp\left(-\frac{\Delta E_2}{k_{\text{B}} T}\right) \tag{3}$$

where g_K, g_Q are the degeneracy factors, m^K, m^Q are the conduction band effective masses for the K and Q band valleys, and ΔE_2 is the energy separation between the K and Q valleys. Equation (2) employs a Fermi–Dirac form for the charge/quantum-capacitance relation, whereas (3) uses the nondegenerate (Boltzmann) weighting to keep a closed-form model; this is consistent with the Taylor-series linearization used from (5) to (6), which recovers the Boltzmann limit.

Rearranging (2), we get

$$V_c(x) = -V_{\text{th}} \ln\left(\exp\left(-\frac{Q_{\text{net}}}{C_{\text{dq}} V_{\text{th}}}\right) - 1\right). \tag{4}$$

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Substituting (4) into (1), we get a uniform charge control model as follows:

$$Q_{\text{net}}(x) - C_t V_{\text{th}} \ln\left(\exp\left(-\frac{Q_{\text{net}}(x)}{C_{\text{dq}} V_{\text{th}}}\right) - 1\right) = -C_t\left(V_g - V_{g0} - V(x)\right). \quad (5)$$

Equation (5) can be solved numerically to find the total mobile charge density. However, numerical solutions are not suitable from a compact modeling perspective as they are computationally too expensive for competitive circuit simulations. Consequently, we make an approximation to the poly-logarithmic term $\ln(\exp(-Q_{\text{net}}(x)/C_{\text{dq}}V_{\text{th}}) - 1)$ to develop our core model.

As an approximation, using a two-term Taylor expansion for the exponential term, we can rewrite the charge control model (which is unified, i.e., valid for all operation regimes) as follows:

$$Q_{\text{net}}(x) - C_t V_{\text{th}} \ln\left(-\frac{Q_{\text{net}}(x)}{C_{\text{dq}} V_{\text{th}}}\right) = -C_t\left(V_g - V_{g0} - V(x)\right). \quad (6)$$

Equation (6) can be solved analytically to find the total mobile charge inside the semiconductor. To do that, we rewrite (6) in the form where the Lambert W function $w\exp(w) = z$ can be applied as follows:

$$\left(-\frac{Q_{\text{net}}(x)}{C_t V_{\text{th}}}\right) \times \exp\left(-\frac{Q_{\text{net}}(x)}{C_t V_{\text{th}}}\right) = \frac{C_{\text{dq}}}{C_t}\exp\left(\frac{V_g - V_{g0} - V(x)}{V_{\text{th}}}\right). \quad (7)$$

Therefore, the total mobile charge inside the semiconductor can be expressed as

$$Q_{\text{net}}(x) = -C_t V_{\text{th}}\, \mathrm{W}\left[\frac{C_{\text{dq}}}{C_t} \times \exp\left(\frac{V_g - V_{g0} - V(x)}{V_{\text{th}}}\right)\right]. \quad (8)$$

Thereby, we have obtained a first approximation for the charge density equation expressed in a closed form using the first branch of the Lambert W function. To improve its accuracy, we introduce a fitting parameter α for the charge density equation, which represents the ideality factor of the subthreshold slope which takes into account the interface trap density and may also be affected by the Schottky barrier

$$Q_{\text{net}}(x) = -C_t V_{\text{th}}\, \mathrm{W}\left[\frac{C_{\text{dq}}}{C_t} \times \exp\left(\frac{V_g - V_{g0} - V(x)}{\alpha V_{\text{th}}}\right)\right]. \quad (9)$$

To calculate the current, we need to calculate the charge densities at the source and drain ends of the channel. The current equation can be expressed as

$$I_{\text{DS}} = \frac{W_L \mu}{L}\int_0^{V_{\text{DS}}} Q(V)\, dV \quad (10)$$

where W_L and L are the width and length of the device, respectively. From (6), we can rewrite dV in terms of dQ as follows:

$$\frac{Q_{\text{net}}(x)}{C_t V_{\text{th}}} - \ln\left(-\frac{Q_{\text{net}}(x)}{C_{\text{dq}} V_{\text{th}}}\right) = -\frac{\left(V_g - V_{g0} - V(x)\right)}{V_{\text{th}}} \quad (11)$$

$$\frac{dQ}{C_t V_{\text{th}}} - \frac{dQ}{Q} = \frac{dV}{V_{\text{th}}} \quad (12)$$

$$dV = \frac{dQ}{C_t} - V_{\text{th}}\frac{dQ}{Q}. \quad (13)$$

Therefore, (10) can be rewritten as

$$I_{\text{DS}} = \frac{W_L \mu}{L}\int_{Q_S}^{Q_D} Q(V)\left[\frac{dQ}{C_t} - V_{\text{th}}\frac{dQ}{Q}\right]. \quad (14)$$

Integrating (14) results in

$$I_{\text{DS}} = \frac{W_L \mu}{L}\left[\frac{Q_D^2 - Q_S^2}{2C_t} - V_{\text{th}}\left(Q_D - Q_S\right)\right] \quad (15)$$

where Q_D and Q_S can be found by substituting $V_D = V_{DS}$ and $V_S = 0$, respectively, in the charge equation, (9).

B. Short-Channel Effects

The core model alone is insufficient for accurately reproducing measurement data from downscaled 2-D FETs, as it does not account for critical physical and SCEs, such as mobility degradation, DIBL, and velocity saturation. To enhance the model's accuracy and better match experimental data, incorporating SCEs is essential, ensuring reliable performance predictions for practical device simulations.

1) Drain-Induced Barrier Lowering: In our investigations, we observed that the 60 nm MoS_2 FET devices exhibit significant DIBL, an electrostatic SCE where the threshold voltage decreases as the drain voltage increases, leading to increased subthreshold current. To accurately capture this behavior, we model DIBL as a linear drain bias-induced shift of the threshold voltage, represented mathematically as follows:

$$V_{g0}(V_{\text{DS}}) = V_{g0} - (\text{DIBL} \times V_{\text{DS}}).$$

This compact representation reflects barrier-lowering by drain coupling and appears as an additive term in the exponent of the unified charge expression, ensuring continuity across regimes. Therefore, the total mobile charge density can be expressed as

$$Q_{\text{net}}(x) = -C_t V_{\text{th}}\, \mathrm{W}\left[\frac{C_{\text{dq}}}{C_t} \times \exp\left(\frac{V_g - V_{g0} + (\text{DIBL} \times V_{\text{DS}}) - V(x)}{\alpha V_{\text{th}}}\right)\right]. \quad (16)$$

2) Mobility Degradation: Mobility degradation is a critical SCE observed in MoS_2 FETs, where the carrier mobility decreases due to scattering at higher vertical electric fields. This degradation impacts the device's drive current and overall performance, particularly in short-channel devices. To model this effect, we introduced a field-dependent mobility term, adjusting the carrier mobility as a function of the effective electric field

$$\mu_{\text{eff}} = \frac{\mu}{1 + \left(\frac{V_g - V_{g0}}{t_f E_0}\right)^{\nu}} \quad (17)$$

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where ν is the mobility effective factor and E_0 is a characteristic electric field. The term $V_g - V_{g0}$ in the effective mobility equation introduces a potential discontinuity, particularly near the threshold voltage where V_g approaches V_{g0}. This discontinuity can lead to nonphysical behavior in the model, affecting the accuracy of the transistor simulations.

To mitigate this issue, a smoothing function is introduced for the overdrive voltage as follows:

$$V_{ov} = \frac{V_{min}}{2}\left(\left(1 + \frac{V_g - V_{g0}}{V_{min}}\right) + \sqrt{(V_{overror})^2 + \left(\frac{V_g - V_{g0}}{V_{min}} - 1\right)^2}\right). \quad (18)$$

Using V_{ov}, μ_{eff} can be rewritten as follows:

$$\mu_{eff} = \frac{\mu}{1 + \left(\frac{V_{ov}}{6E_0}\right)^{\nu}}. \quad (19)$$

3) Velocity Saturation: To incorporate velocity saturation into the model, V_{DS} is replaced with an effective drain-to-source voltage V_{DSeff}, and a V_{DS} related term is added to the current equation. V_{DS} is represented as follows:

$$V_{DSeff} = \frac{V_{DS}}{\left[1 + \left(\frac{V_{DS}}{V_{DSsat}}\right)^m\right]^{\frac{1}{m}}} \quad (20)$$

where m is a parameter that gives the transition from the linear to the saturation regime and V_{DSsat} is expressed as follows:

$$V_{DSsat} = \frac{\left(V_{ov} \times \frac{2v_{sat}L}{\mu_{eff}}\right)}{V_{ov} + \frac{2v_{sat}L}{\mu_{eff}}}. \quad (21)$$

Equation (21) is a smoothing function that blends the two limits: the electrostatic one $V_{DSsat} \approx V_{ov}$ and the velocity-limited one $V_{DS_sat} \approx 2\, v_{sat}L/\mu_{eff}$, where V_{ov} is the smoothing function to the overdrive voltage used to ensure continuity and prevent any discontinuities, v_{sat} is the saturation velocity, μ_{eff} is the effective mobility and L is the channel length. The drain current, therefore, can be expressed as follows:

$$I_{DS} = \frac{W_L\mu_{eff}}{L}\left[\frac{Q_D^2 - Q_S^2}{2C_t} - V_{th}(Q_D - Q_S)\right] \times \left(\frac{1}{1 + \frac{V_{DSeff}}{E_{sat} \times L}}\right). \quad (22)$$

4) Channel Length Modulation: We also account for CLM, an SCE where the effective channel length decreases with increasing drain voltage (due to the increase of the channel length portion where the carrier velocity is saturated), leading to a slight increase in current at higher drain voltages. To model this, we included a CLM correction factor in the drain current equation, enhancing the accuracy of our model for short-channel devices. The total current equation can be expressed as follows:

$$I_{DS} = \frac{W_L\mu_{eff}}{L}\left[\frac{Q_D^2 - Q_S^2}{2C_t} - V_{th}(Q_D - Q_S)\right] \times \left(\frac{1}{1 + \frac{V_{DSeff}}{E_{sat} \times L}}\right) \times (1 + \lambda V_{DS}) \quad (23)$$

where λ is the CLM parameter.

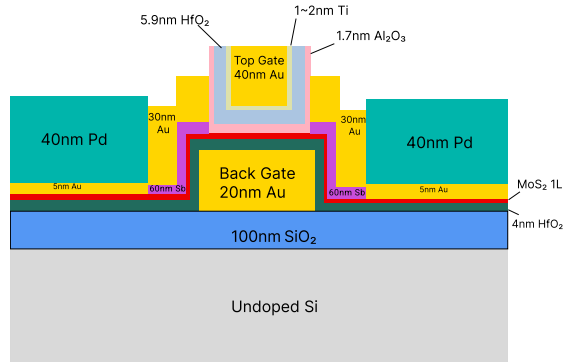

Fig. 1. Cross-sectional view of the double-gated MoS_2 FET, showing the top gate oxide stack made of Al_2O_3 and HfO_2, with the local back gate oxide consisting of HfO_2.

III. Results and Discussion

Our model was validated using experimental data from MoS_2 double-gated FETs fabricated at Intel Foundry Technology Research Labs and measured at TU Vienna [19]. The fabrication involved industry-compatible processes, starting with the formation of a SiO_2 insulating layer thermally oxidized on an undoped Si wafer. This was followed by the deposition of a 4 nm HfO_2 layer (local gate insulator) via atomic layer deposition (ALD). A MoS_2 layer grown via metal-organic chemical vapor deposition (MOCVD) was then transferred to form the transistor channel [20]. Source and drain contacts were patterned using an Sb/Au metal stack, chosen to minimize Fermi-level pinning and ensure low-resistance, ohmic contacts [2], [21] [22], [23]. Sb was specifically selected over Bi due to its superior thermal stability [22]. Finally, a top-gate-stack was deposited, consisting of 1.7 nm Al_2O_3 and 5.9 nm HfO_2 for the local back-gate devices. A schematic of the device is presented in Fig. 1. In the considered double-gate structure, the top gate is actively biased, while the bottom gate is kept grounded. The grounded gate contributes to the overall electrostatics of the channel through capacitive coupling, effectively shifting the threshold voltage. Therefore, the model formulation is based on the top-gate bias, and the influence of the grounded gate is implicitly accounted for in the extracted threshold voltage. The model accurately reproduces the experimental I–V characteristics and effectively captures SCEs, such as mobility degradation and velocity saturation, demonstrating very good agreement with the measured data.

The results shown in Fig. 2 are for double-gated devices with channel lengths of 500, 200, and 60 nm. All the devices have the same channel width (W_L) of 1 μm. As shown in Fig. 2, mobility degradation is evident in the transfer curves for all devices, as reflected in the decreasing slope of the current in the linear region of the characteristics as the gate voltage increases. This reduction likely arises from increased scattering due to interface roughness and imperfections. Moreover, for the 60 nm channel length, we also observe band-to-band tunneling (BTBT) at high V_{DS}, which increases the leakage current. However, this effect is not captured by the present model; comprehensive leakage-current modeling (including BTBT) is beyond the scope of this article and will be addressed

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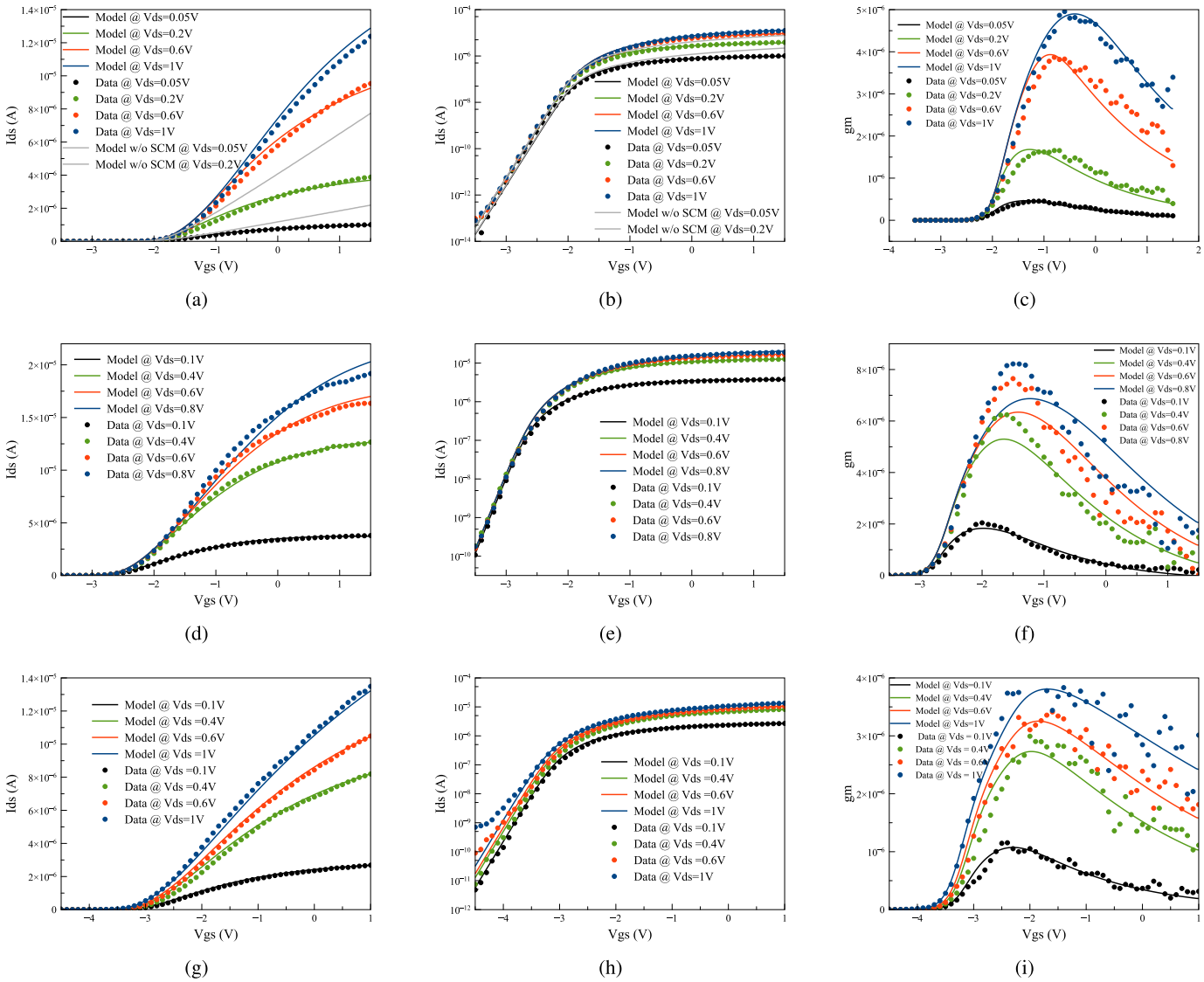

Fig. 2. Validation of the compact model against experimental data for double-gate MoS_2 FET with L = 500 nm (first row), L = 200 nm (second row), and L = 60 nm (third row). All devices shown have channel width W = 1 μm. (a) Linear transfer characteristics for L = 500 nm (I_{ds}–V_{gs}) for V_{ds} = 0.05, 0.2, 0.6, 1 V, where the gray line represents the model results when mobility degradation is ignored, illustrating the importance of including this effect. (b) Log-scale transfer characteristics. (c) Transconductance (g_m = dI_{ds}/dV_{gs}), model versus measurement. (d) Linear transfer characteristics for L = 200 nm for V_{ds} = 0.1, 0.4, 0.6, 0.8 V. (e) Log-scale transfer characteristics. (f) Transconductance (g_m = dI_{ds}/dV_{gs}), model versus measurement. (g) Linear transfer characteristics for L = 60 nm for V_{ds} = 0.1, 0.4, 0.6, 1 V. (h) Log-scale transfer characteristics. (i) Transconductance (g_m = dI_{ds}/dV_{gs}), model versus measurement.

in future developments. In [14], it was shown that when the film has a high density of states, mobility should be modeled as a power-law function of (V_g–V_{g0}). However, this was not necessary to fit the I–V characteristics of the technology targeted in this work. Additionally, electrostatic SCEs lead to an increase in subthreshold swing (SS) and a reduction of V_{g0} as channel length decreases. DIBL is only significant at very short-channel lengths. For the longest channel device, the slight overestimation of I_D at high V_g–V_{DS} stems from our choice to keep key high-field parameters (e.g., v_{sat} and the soft-saturation smoothing) and the vertical-field mobility law μ_{eff} length-invariant to avoid overparameterization. In the long-channel limit, where the intrinsic channel drop dominates and SCEs are weak, any overestimation in μ_{eff} law becomes more visible, yielding a modest overestimate at high bias. Introducing a weak L-dependence for the soft-saturation onset or a slightly stronger μ-degradation would remove this bias, but at the expense of additional parameters; we have favored the channel length independent formulation. We will consider these points in future extensions of the model by revisiting the length-invariant assumptions and, where warranted, introducing minimal L-dependence or refined high-field and mobility terms, while maintaining a compact parameter set.

Fig. 3 shows the output characteristics for the same devices. As seen in Fig. 3, the output drain current I_{ds} increases with the applied drain–source voltage V_{ds} across all channel lengths (500, 200, and 60 nm), indicating the presence of CLM. The slope of the I_{ds} curves becomes more pronounced as L decreases, due to the increase of the fraction of the channel length where the velocity is saturated. We treat λ as a lumped

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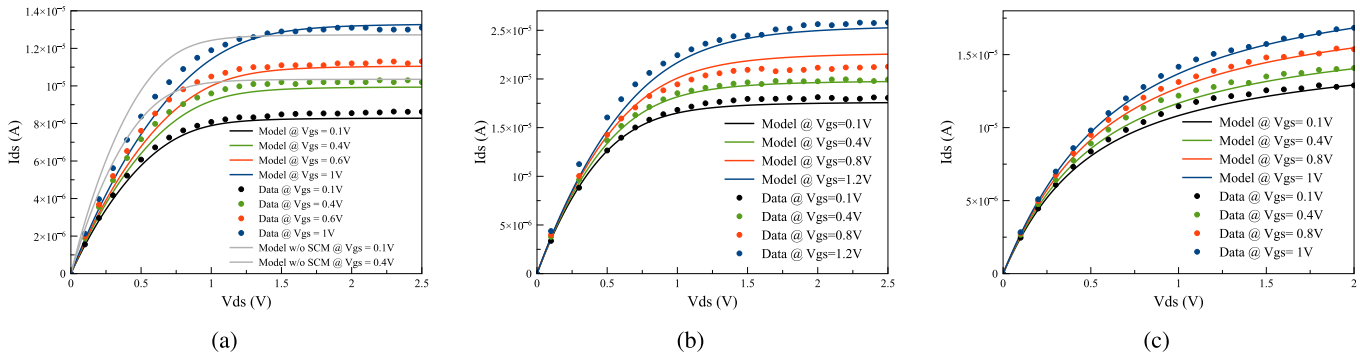

Fig. 3. Validation of the output characteristics (I_{ds}–V_{ds}) of the compact model against experimental data for a double-gated device at different gate voltages. (a) Output characteristics for L = 500 nm, where the gray line represents the model results when mobility degradation is ignored, illustrating the importance of including this effect. (b) L = 200 nm. (c) L = 60 nm.

TABLE I
INPUT PARAMETERS FOR THE 2-D MONOLAYER FET MODEL

Parameter	Symbol	L = 500 nm	L = 200 nm	L = 60 nm
Width	W (μm)	1	1	1
Mobility	μ (cm^2/Vs)	150	140	43.5
Threshold Voltage	V_{g0} (V)	-1.6	-2.28	-2.6
SS ideality factor	α (unitless)	4	4.3	5.3
Velocity saturation factor	v_{sat} (m/s)	2.07×10^4	2.67×10^4	2.6×10^4
Mobility effective factor	ν (unitless)	1.3	1.65	1.3
Linear-Saturation Transition factor	m (unitless)	5.4	3.6	1.4
Channel Length Modulation (CLM)	λ (1/V)	0.001	0.001	-0.035
DIBL	DIBL (1/V)	0	0	0.2

high-field fitting factor; a small negative value (seen only at $L = 60$ nm) compensates a slight overprediction of the high-V_{DS} slope once DIBL and soft-saturation are included and does not imply a physical sign reversal of CLM. The calculated effective mobility decreases with reducing L due to the impact of series resistance. Series/access resistances are critical in 2-D FETs, as they introduce a nonnegligible voltage drop outside the intrinsic channel at high current (e.g., $V_{DS,ch} = V_{DS} - I_{DS}(R_s + R_d)$), thereby reducing the measured drain current relative to a channel-only model. In this work, we keep the mobility law intrinsic and defer an explicit, bias- and temperature-dependent extraction of R_s and R_d to future studies with a wider set of channel lengths. Table I shows the input parameters for the model. All model parameters are attributed to control by the top gate, while the bottom gate influences only the threshold voltage V_{g0}. Within this framework, the saturation velocity is a material property; therefore, we treat v_{sat} as an effective high-field fitting parameter, and the small variations observed across L are interpreted as fitting variance due to parameter correlation rather than a structural length dependence. Parameter extraction proceeds in three steps. First, from long-channel devices, low-V_{DS} transfer curves, we extract the threshold voltage V_{g0} and the SS via the ideality factor α, with SS = $\alpha(kT/q)\ln 10$. Second, we calibrate the mobility parameter values in a staged, physics-based manner: we first extract the low-field mobility from long-channel, small-V_{DS} data using the linear-region transconductance, we then lock μ and tune the degradation law $\mu_{eff}(\nu, E_0)$ so that the model is anchored in the low-V_g regime and subsequently reproduces the high-V_g roll-off in a physics-consistent manner. Third, keeping V_{g0}, α, and the mobility parameter values fixed, we fine-tune the remaining high-field parameters, velocity saturation (or critical field), DIBL, and CLM, to achieve a simultaneous fit of transfer and output characteristics across biases and gate lengths, while maintaining physically consistent trends.

IV. CONCLUSION

In this work, we present a dc compact model for short-channel MoS_2 FETs. Based on a unified charge control model developed initially for long-channel devices, the proposed model incorporates SCEs, with the channel charge sheet density expressed through the Lambert W function. The current is calculated using the charge density expressions at the source and drain ends of the channel. The model demonstrates excellent agreement with experimental data and accurately reproduces the behavior of FETs for channel lengths down to 60 nm. Ultimately, this model facilitates more robust and efficient circuit simulations, enabling better design and optimization of systems utilizing double-gated MoS_2 FETs.

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